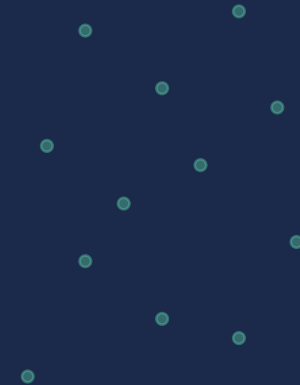


InsideInsight



Strategic Airbnb Optimization via Big Data & Agentic AI

Team 9 · MSBA 6331 Big Data Analytics · Carlson School of Management, University of Minnesota

Bhavisha Chafekar · Jyothirmai Sri Peesapati · Phoenix Ferrari · Stephen Weiler · Tzu-Yu Chen

1,955,039

data rows

35,957

listings

5 cities

US markets

50,000

reviews NLP

P1

Data Ingestion

Inside Airbnb CSV
5 cities · Bronze/Silver/Gold
Delta Lake on Databricks

gold_listings
gold_calendar_daily
gold_reviews

P2

Pricing Analytics

IQR outlier detection
90-day rolling occupancy
Competitive score 0–100

gold_features
1,955,039 rows

P3

NLP Reviews

VADER sentiment
7-topic extraction
Top praise + complaint

gold_nlp_features
50,000 reviews scored

P4

AI Agent

FAISS vector search
LangGraph 3-node pipeline
LLaMA 3.3 70B via Groq

action_plan()
Grounded recs

P5

Dashboard

Streamlit 4 pages
URL lookup · Filters
Live demo app

InsideInsight
Streamlit app

Key formula

Revenue upside = price gap (\$) × occupancy days

Example: \$22 gap × 255 days = \$5,610/year recoverable

Competitive score = 30% price + 25% amenity + 25% reviews + 20% occupancy

Technology stack

Apache Spark · Databricks · Delta Lake · PySpark
VADER Sentiment · SentenceTransformers · FAISS
LangGraph · Groq LLaMA 3.3 70B · Streamlit · Python

What makes us different

Grounded AI — every recommendation cites exact stats
Hallucination validator on all agent outputs
1.95M rows on Databricks — not a spreadsheet
Open data (Inside Airbnb CC4.0) — fully reproducible

Step 1

IQR Outlier Detection

- Removes outliers with Q3 + 1.5×IQR upper bound per city

clean_listings

Step 2

Occupancy Modeling

- Overall 90-day rolling window
- Calculate peak month per neighborhood
- From calendar data

occ_90d_df · peak_months

Step 3

Competitive Score ranging from 0–100

- Price gap % (30%)
- Amenity score (25%)
- Review rank (25%)
- Occupancy rank (20%)

listings_scored

Step 4

gold_features Delta Table

- Join peak season data
- Format & round columns
- Result power product dashboard

workspace.default.gold_features

Key Outputs

~1.95 Million

rows → gold_features

14 engineered features

Competitive Score Formula

Score = Price Gap×0.30 + Amenity×0.25 + Review Rank×0.25 + Occ Rank×0.20

> 80 Market Leader

40–80 Mid-tier improvements

< 40 Fixer-Upper

Technology Used

PySpark

Window functions, percent_rank()

Delta Lake

gold_features overwrite schema

Dashboard Demo — Sample Output

Chicago listing #20406582 · Albany Park · Entire home

Nightly price

\$98

vs \$120 median

Price gap

-18.3%

Underpriced

Occupancy

255 days

69.9% rate

Annual revenue

\$24,990

Estimated actual

Revenue upside

+\$5,610

\$22 × 255 days

You are underpriced by \$22/night — costing an estimated \$5,610/year in lost revenue

Formula: Price gap (\$22) × Occupancy days (255) = \$5,610/year recoverable · Peak demand: August · Amenity score: 43/100

4 dashboard pages

Dashboard

Pricing gap · occupancy · revenue upside · NLP sentiment · reviews

Pricing simulator

Drag slider → see revenue change. Revenue curve + neighbourhood histogram

Health check

Traffic light scorecard across 5 dimensions + neighbourhood leaderboard

AI advisor

LangGraph → FAISS → LLaMA 3.3 70B → 3 grounded recommendations

Review intelligence + AI agent

Sentiment

0.78

Very Positive

Reviews

521

VADER scored

Top praise

Location

Most mentioned

Top complaint

Cleanliness

Most mentioned

LangGraph AI Agent

retrieve context + FAISS comparables → LLaMA 3.3 70B analysis → 3 grounded recommendations

Every recommendation cites exact source stats · Hallucination validator on all outputs

Understanding Guest Reviews at Scale

3 Million

guest reviews analyzed
across 5 cities

For every listing, we extract:

Sentiment score

Positive or negative overall?

Top praise

What guests love

Top complaint

What's hurting the rating

How it works

1

Clean & Filter

We remove noise — keep only English reviews, strip junk text

 *PySpark · langdetect*

2

Score the Sentiment

An AI model reads each review and scores how positive or negative it is

 *DistilBERT transformer · HuggingFace · pandas_udf*

3

Find the Themes

We detect what topics keep coming up — WiFi, cleanliness, check-in, location

 *BERTopic · 100K-review sample · 13 topic clusters*

4

Power the AI Agent

Results power the AI agent — e.g. 'WiFi flagged in 42% of your reviews'

 *gold_nlp_features Delta table → LangGraph agent (Phase 4)*